

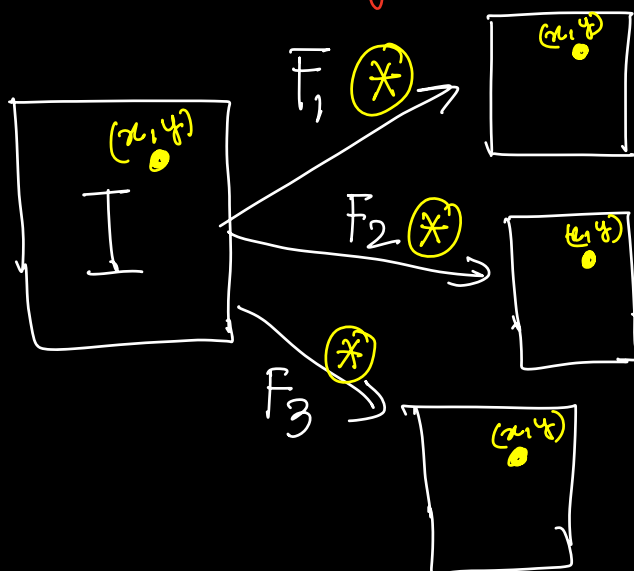
CONVOLUTIONAL NEURAL NETWORK

This lecture is adapted from "How Convolutional Neural network works by Brandon Rohrer".

(available on Youtube)

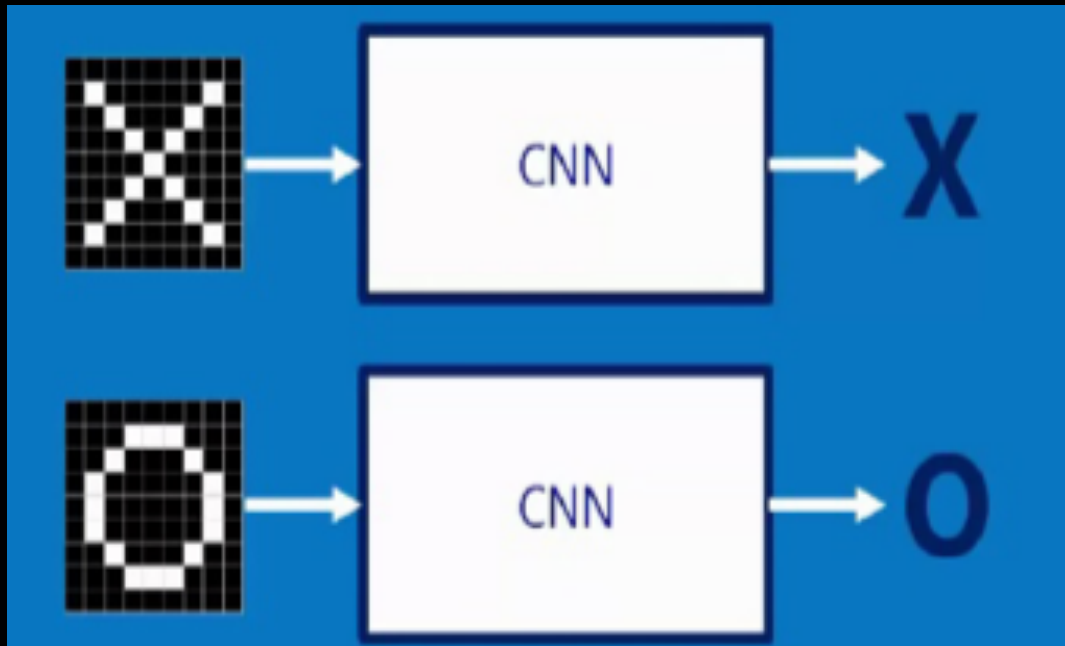
- ⊗ Visual interview of an image $I(m, n)$.
- ⊗ Visually relevant questions, that can be useful for obtaining discriminative feature (Response), need to be asked. (They are encoded within a filter)

- What is your directional ^{gradient} $F(K \times K)$
- Do you belong to a line like $(/)$ or $(\backslash)$ (1)?
- Do you have a pattern like $(+)$ or (X) ----?
- Do you belong to red blob / or similar

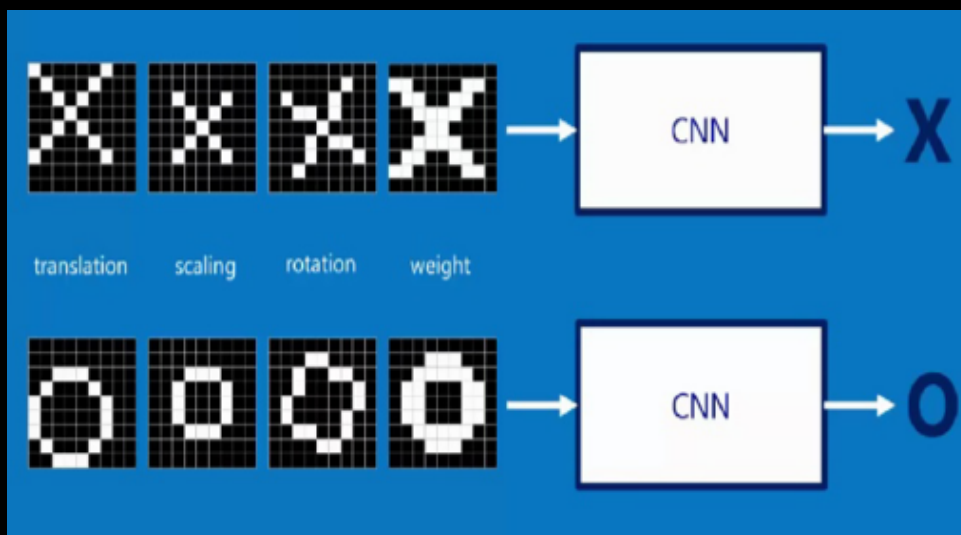


$I(x, y)$ is the pixel under consideration
 $\Downarrow$
 $F_1(x, y), F_2(x, y), F_3(x, y) \Rightarrow$ They are its responses for the specifically designed question w.r.t the objective fn.

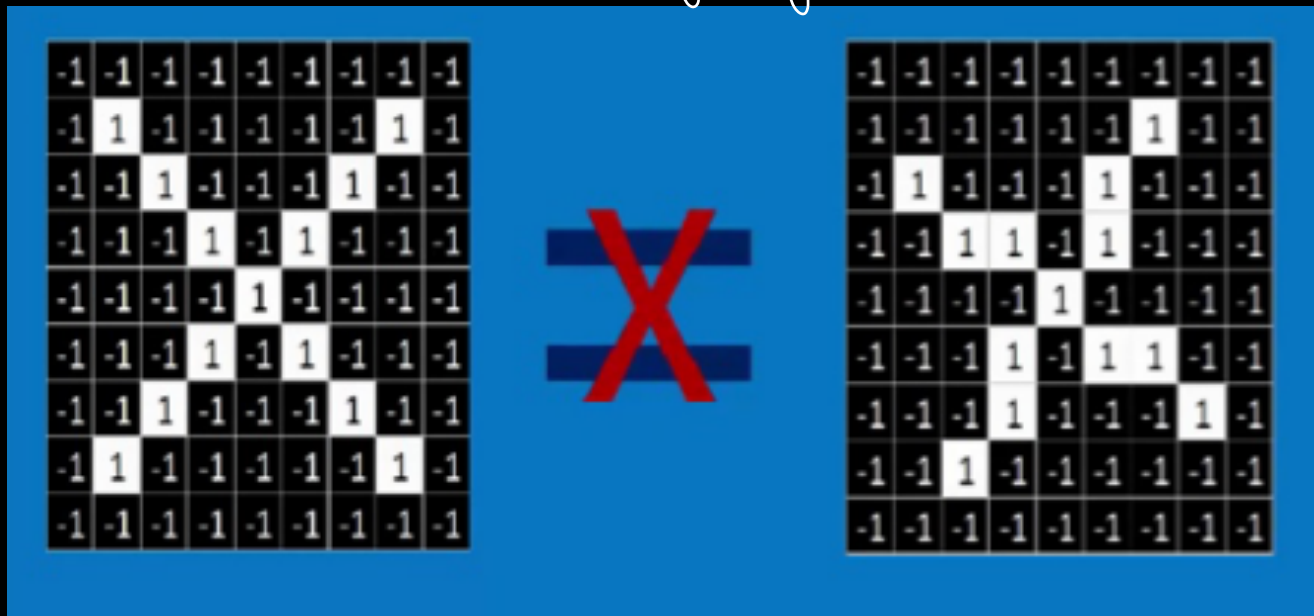
Let us talk about a problem



Design a CNN network that
can classify/differentiate b/w $\textcircled{X}$ &
 $\textcircled{0}$ $\Rightarrow$ Image classifier.



These variations really makes this problem very difficult to solve.



Q1 Both are same or different

Q2 Both of them are (X) or (O)

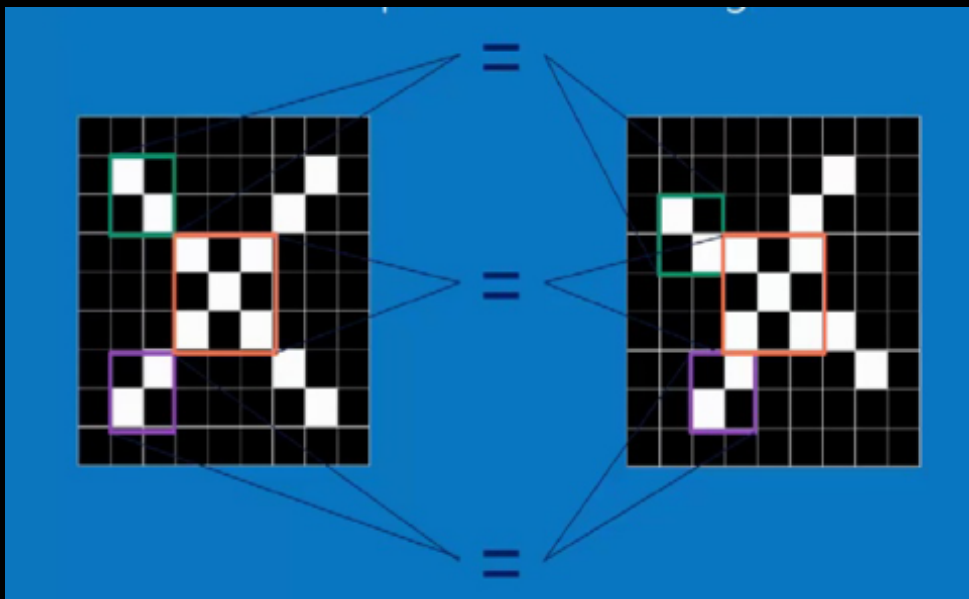
⇒ Q1 & Q2 Both are
2 Completely different problems.

Q1 $\Rightarrow$ Metric learning

Q2 $\Rightarrow$ Classification

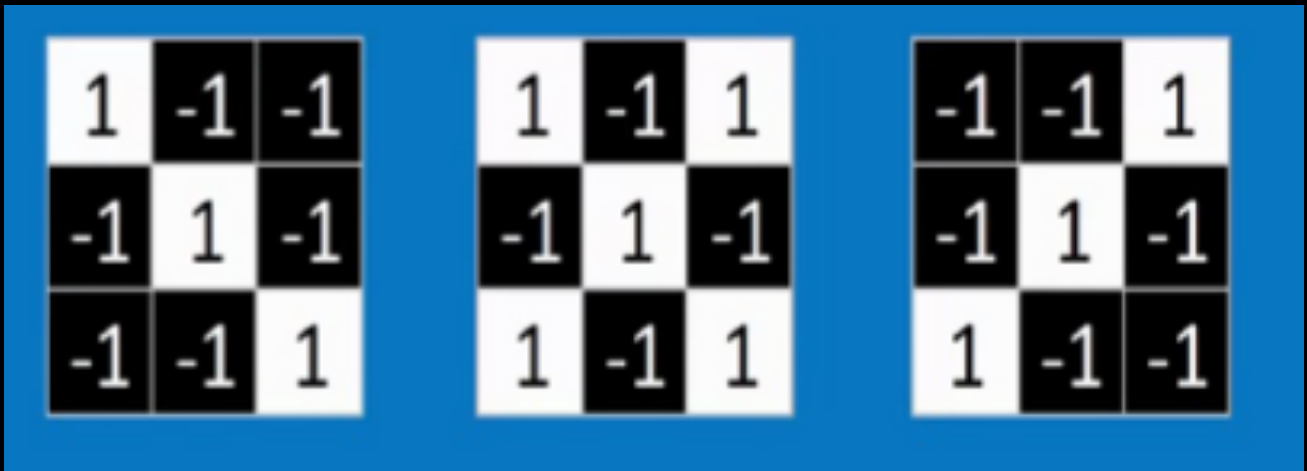
-1	-1	-1	-1	-1	-1	-1	-1	-1
-1	X	-1	-1	-1	-1	X	X	-1
-1	X	X	-1	-1	X	X	-1	-1
-1	-1	X	1	-1	1	-1	-1	-1
-1	-1	-1	-1	1	-1	-1	-1	-1
-1	-1	-1	1	-1	1	X	-1	-1
-1	-1	X	X	-1	-1	X	X	-1
-1	X	X	-1	-1	-1	-1	X	-1
-1	-1	-1	-1	-1	-1	-1	-1	-1

One can see
that at so many
places they are
different.
16/81 (mis-match)
High degree of
dis-similarity.



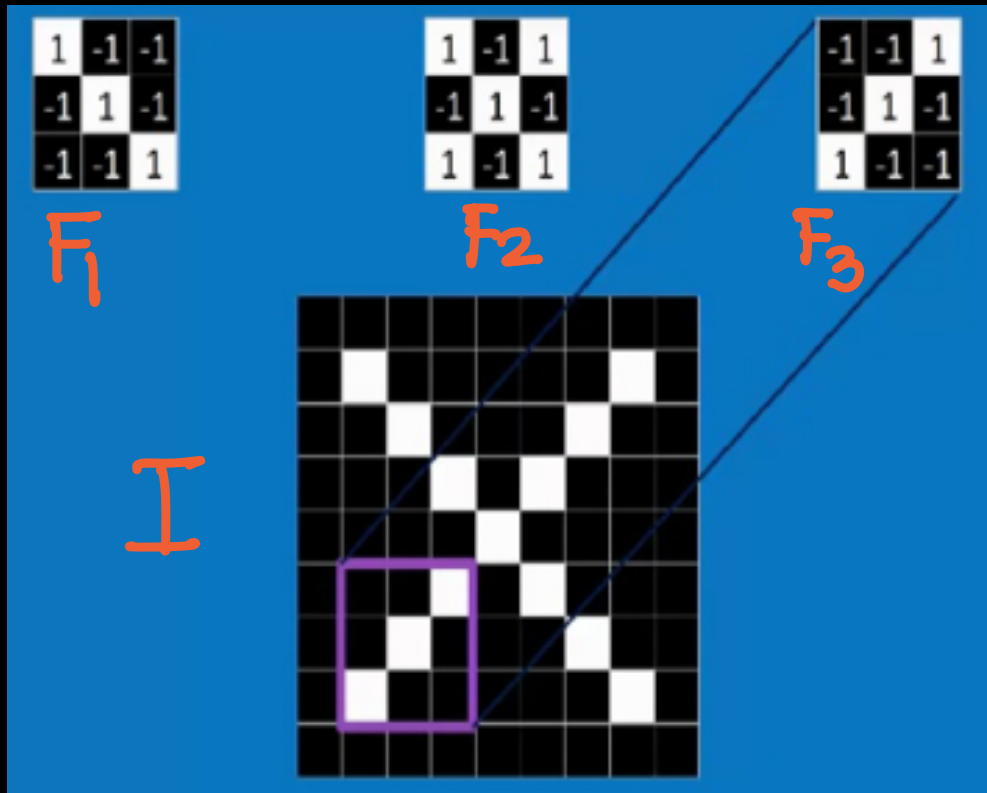
Partwise
they are
exact match.

Ⓐ CNN can help you to see the image in parts.



These can be useful questions that need to be asked if we need to detect an $\otimes$.

⊛ Right now we assume some oracle is providing us that
 $\Rightarrow$ But similar to FCN, they are also learned via. Back Propagation.

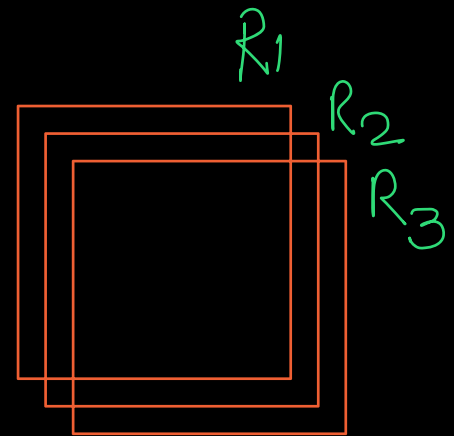


These filters have some part based questions embedded into it.

$$I \otimes F_1 \Rightarrow R_1$$

$$I \otimes F_2 \Rightarrow R_2$$

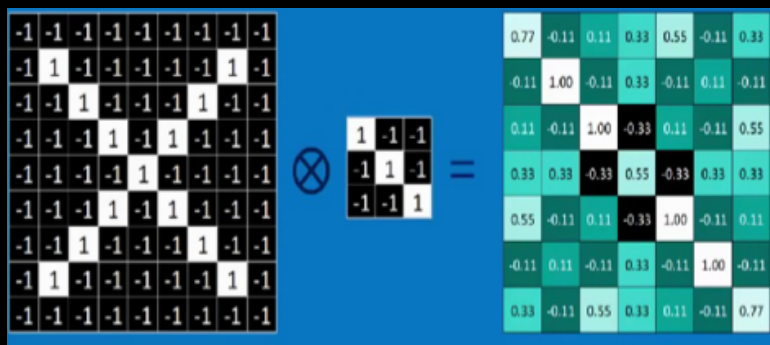
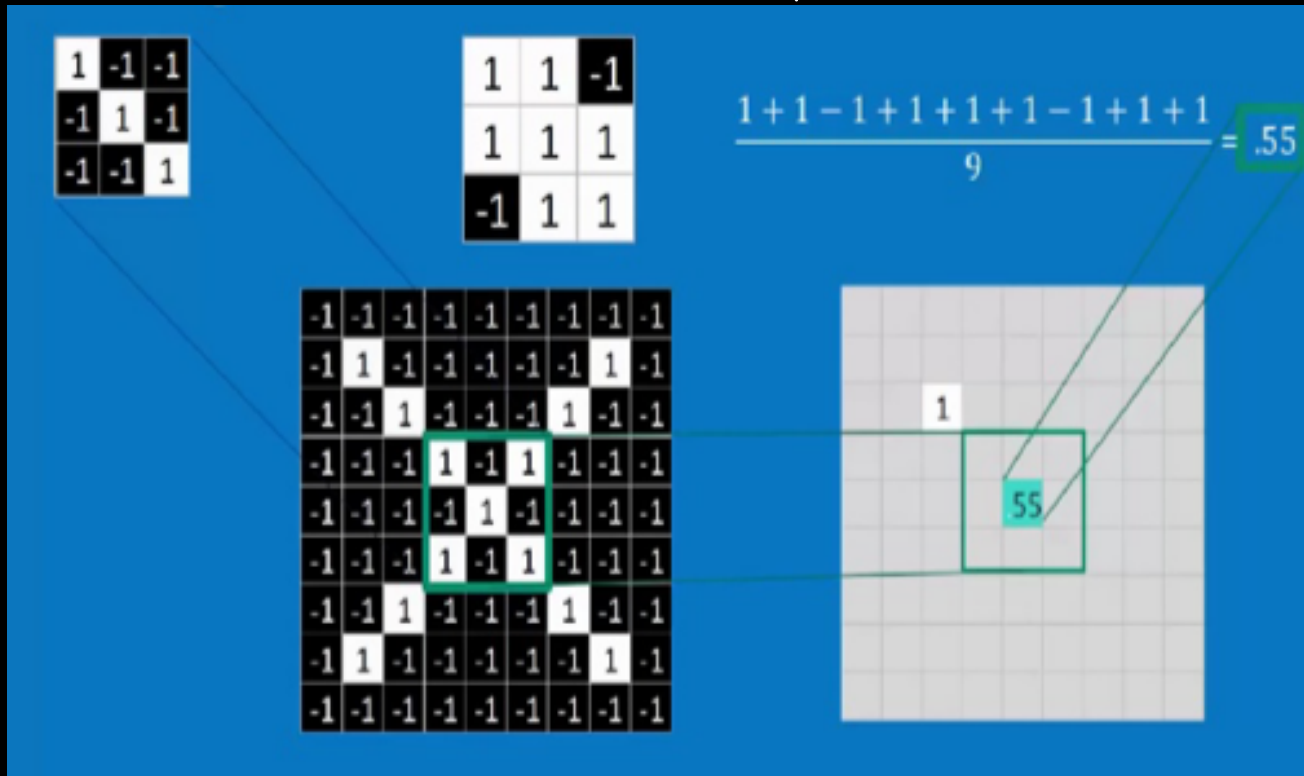
$$I \otimes F_3 \Rightarrow R_3$$



Convolve image using these filters to get the response map.

(Response Map) or extracted features.

Basic Convolution operation



One can understand the response obtained.

All these filters placed in one layer $\Rightarrow$ CNN layer.



① One 1 filter $\Rightarrow$ generates one Response map.

② One CNN layer may have $N = 128, 256$ such filters of size may be (3×3) & (5×5)

$\Rightarrow \# \text{ parameters} = N \times 3 \times 3$

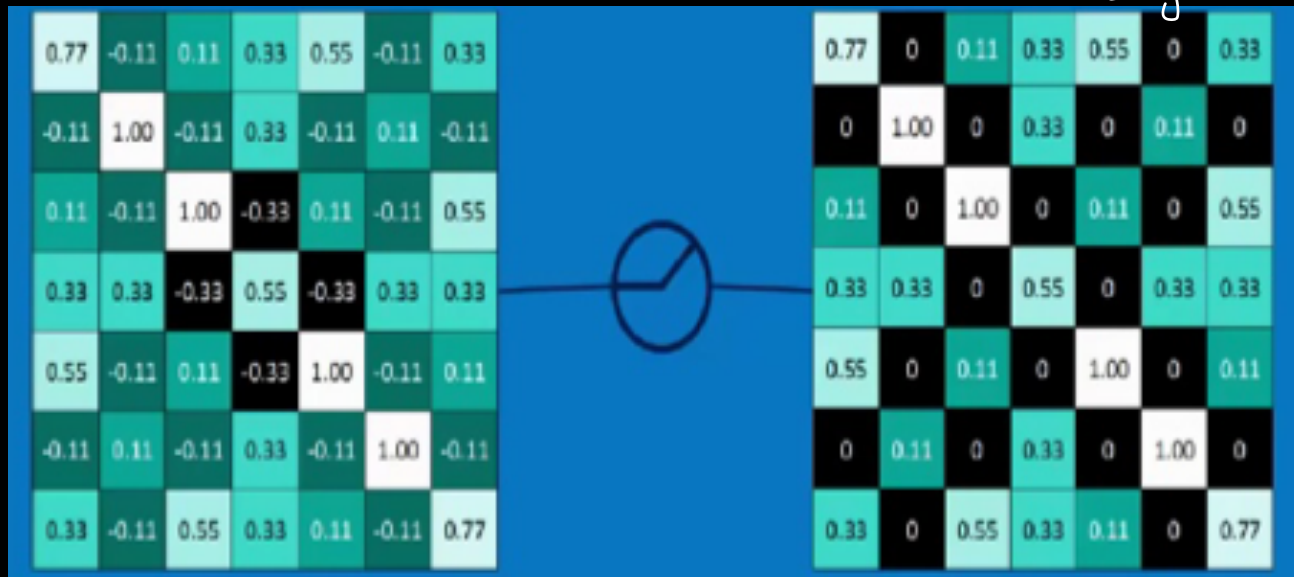
OR $N \times 5 \times 5$

Question:

OR $(N \times K \times K)$.

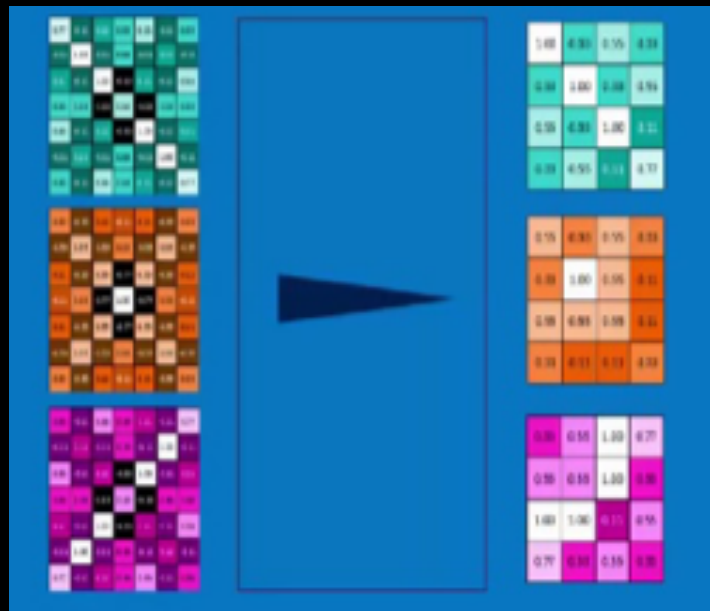
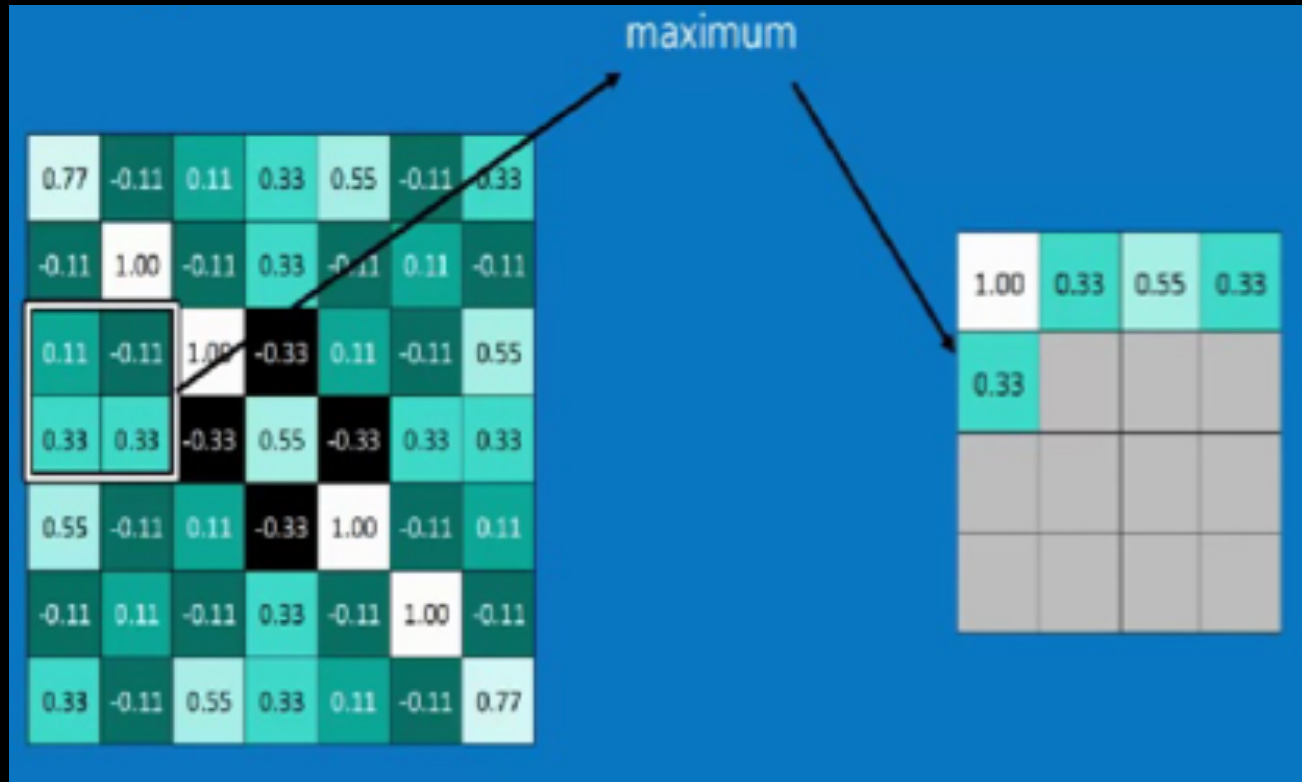
Given an input image of size 28×28 , if we apply a CNN layer with (64) (3×3) filters then what is the O/P dimensions.

These response maps are
 activated using Sigmoid,
 tanh or ReLU. (-Normalizing)
 - Adding
 non-linear fn.



(B)

Pooling : Feature
Size reduction



⇒ This is
Basically
the working of
2x2 pooling
over response map.

⑦ (*) Pooling is important in order to manage the feature size.

⑧ (*) Aggregating the features so as one can move from local to a global features.

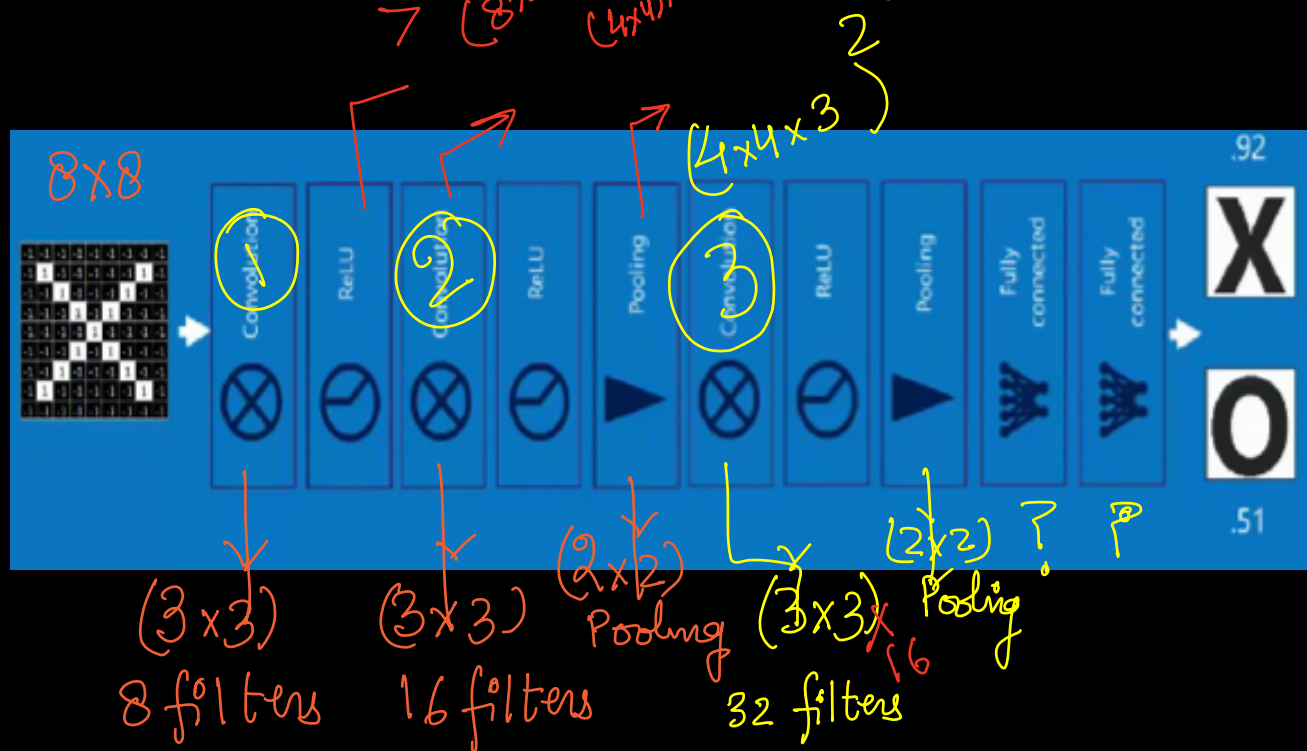
⑨ (*) Also provide some amount of robustness against Rotation/translation.

⑩ (*) 2×2 pooling is common and will reduce the h & w by (2).

⑪ (*) Several times if images are very big we need to pool after each Conv, but when feature becomes small $\Rightarrow$ less frequent Pooling.

C Deep Stacking

⇒ This will give us our first CNN based Neural Net.



(*) CNN exploits spatial relationship that got disturbed by FCN & Cannot be utilized in 2D.

CNN Can ask
(2D) Questions

|| FCN was asking
(1D) questions

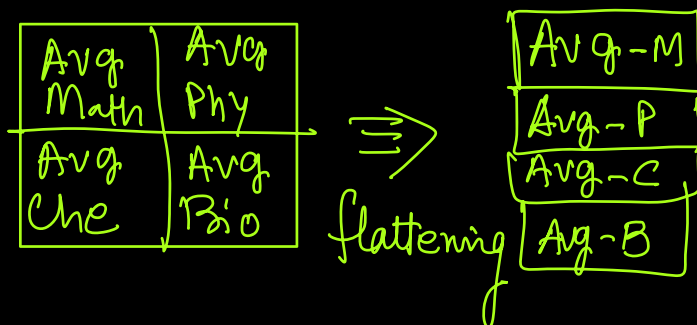
⑧ Pooling allows aggregation by feature reduction (Max-Pool)

local $\Rightarrow$ Global estimates

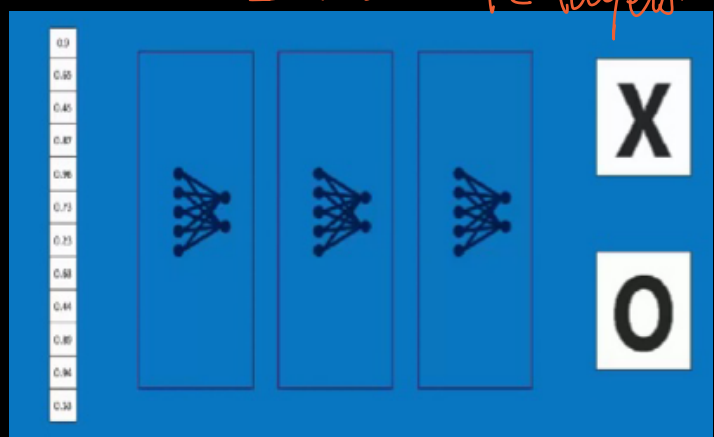
$\hookrightarrow$ At some point (feature size)
global estimates contains the
global image statistics like:

average of (Color, edges,

They will not have spatially important relationship
Variation of certain type, Average occurrence of a pattern)



Flatten & apply
small FC layers.



Hyper-parameters

(a) Convolutional layers

- A.F
- # of filter
- Size of filter
- Stride in x & y

(b) Pooling

- Window Size
- Window Stride in x & y

(c) Full Connected

- # of layers
- # of Neurons in each layer.

